

Graphing Quadratic Expressions (Part IV)

Classifying Numbers

Real Number: A number that is not imaginary (imaginary numbers are numbers that are written with a factor of $i = \sqrt{-1}$, which you will learn later in this course).

Integer: A whole number $(-3, -2, -1, 0, 1, 2, 3, \dots)$.

Natural Number: A whole number greater than zero $(1, 2, 3, 4, \dots)$.

Non-Terminating Decimal: A decimal that continues without stopping $(\frac{1}{3} = 0.3333\dots$ or $0.\overline{3})$.

Non-Repeating Decimal: A non-terminating decimal whose digits display no pattern, i.e. the digits don't repeat $(\sqrt{3} = 1.732050807\dots)$.

Rational Number: A number that can be written as a fraction with an integer numerator and an integer denominator $(\frac{2}{3}, -4, 7, \frac{1}{4})$.

Irrational Number: A number that cannot be written as a fraction with an integer numerator and an integer denominator $(\pi, \sqrt{5}, \sqrt{3}/2)$.

Irrational numbers are non-terminating, non-repeating decimals $(\sqrt{3}, \pi, \sqrt{7}/5)$.

Some rational numbers can be written as non-terminating, repeating decimals $(\frac{5}{6} = 0.83333\dots$ or $0.8\overline{3})$.

Identifying Irrational Numbers

Circle the irrational number in each list below. You may use a calculator to convert to decimal form if helpful.

Example I) $3.14, 8, \frac{1}{2}, \sqrt{2}, 0.\overline{7}$

Example II) $0.0\overline{6}, 0.\overline{348}, \frac{4}{7}, 2\pi, (\sqrt{3})^2$

Example III) $\sqrt{\frac{1}{9}}, -2.\overline{291}, 3^2, \frac{1}{\sqrt{6}}, \frac{1}{3}$

Example IV) $\sin 62^\circ, 5\frac{2}{7}, \frac{\pi}{\pi}, -8.\overline{8}, \frac{22}{23}$

Example V) $-7.33331, \sqrt{10}, -9, \cos 60^\circ, \frac{\sqrt{5}}{\sqrt{5}}$

Example VI) $\tan 45^\circ, \frac{7}{\pi}, -18.0064, \frac{3}{2}, 0$

Example VII) $-\frac{1}{34}, \frac{0}{6}, -4.7, 0.\overline{345}, \pi^2$

Example VIII) $\sqrt{4}, -\frac{731}{999}, \frac{6}{\tan 37^\circ}, \sqrt{100}, \frac{8}{65}$

Graphing Quadratic Expressions

Standard Form	Intercept Form	Vertex Form
$y = ax^2 + bx + c$	$y = a(x-m)(x-n)$	$y = a(x-h)^2 + k$

Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x-value of the vertex is $\frac{m+n}{2}$.

* Intercept form is not commonly used to graph a quadratic expression if the corresponding graph does not move through the x-axis.

Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then (h,k) is the vertex.

Graph each quadratic expression. Show the coordinates of the vertex, y-intercept, and x-intercepts (if there are any). Draw the line of symmetry (as a dotted line) and write its equation.

① $3(x+1)^2 - 6 = y$

$a=3, h=-1, k=-6$

I) Vertex

$V_x = h = -1, V_y = k = -6$

$V(-1, -6)$

II) y-Intercept

$x=0$

$3(0+1)^2 - 6 = y$
 $-3 = y$

$(0, -3)$

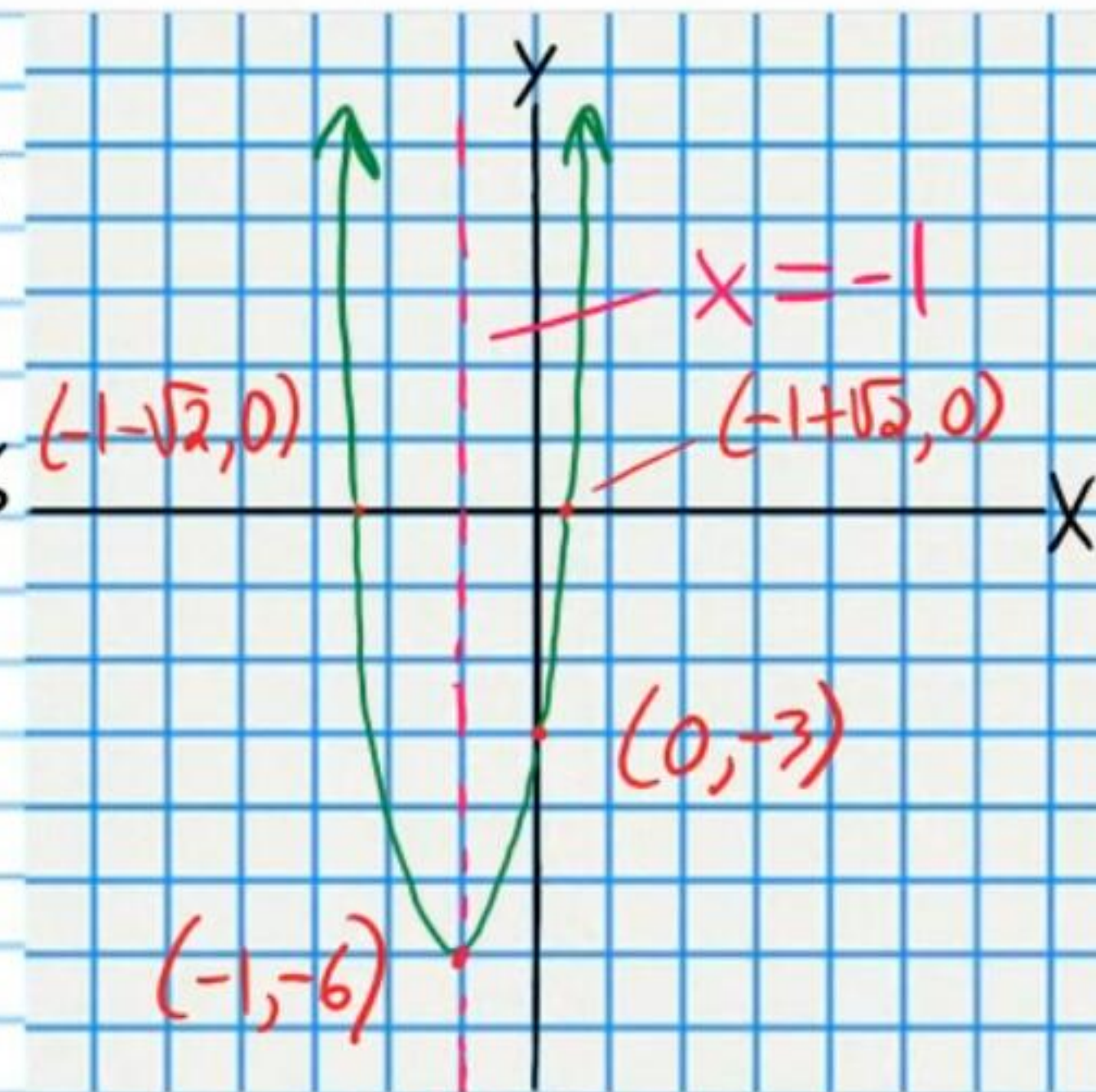
III) x-Intercepts

$y=0$

$3(x+1)^2 - 6 = 0$

$3(x+1)^2 = 6$

$(x+1)^2 = 2$



$\sqrt{(x+1)^2} = \pm\sqrt{2}$

$x+1 = \pm\sqrt{2}$

$x = -1 \pm \sqrt{2}$

$x = -1 + \sqrt{2}$ or $x = -1 - \sqrt{2}$

$(-1 + \sqrt{2}, 0)$ & $(-1 - \sqrt{2}, 0)$

$x \approx -1 + 1.414$ or $x \approx -1 - 1.414$

$x \approx 0.414$ or $x \approx -2.414$

$$\textcircled{2} y = -2(x-2)^2 + 6$$

$$a = -2, h = 2, k = 6$$

I) Vertex

$$V_x = h = 2, V_y = k = 6$$

$$V(2, 6)$$

II) Y-Intercept

$$x = 0$$

$$y = -2(0-2)^2 + 6$$

$$y = -2(-2)^2 + 6$$

$$y = -2(4) + 6$$

$$y = -2$$

$$(0, -2)$$

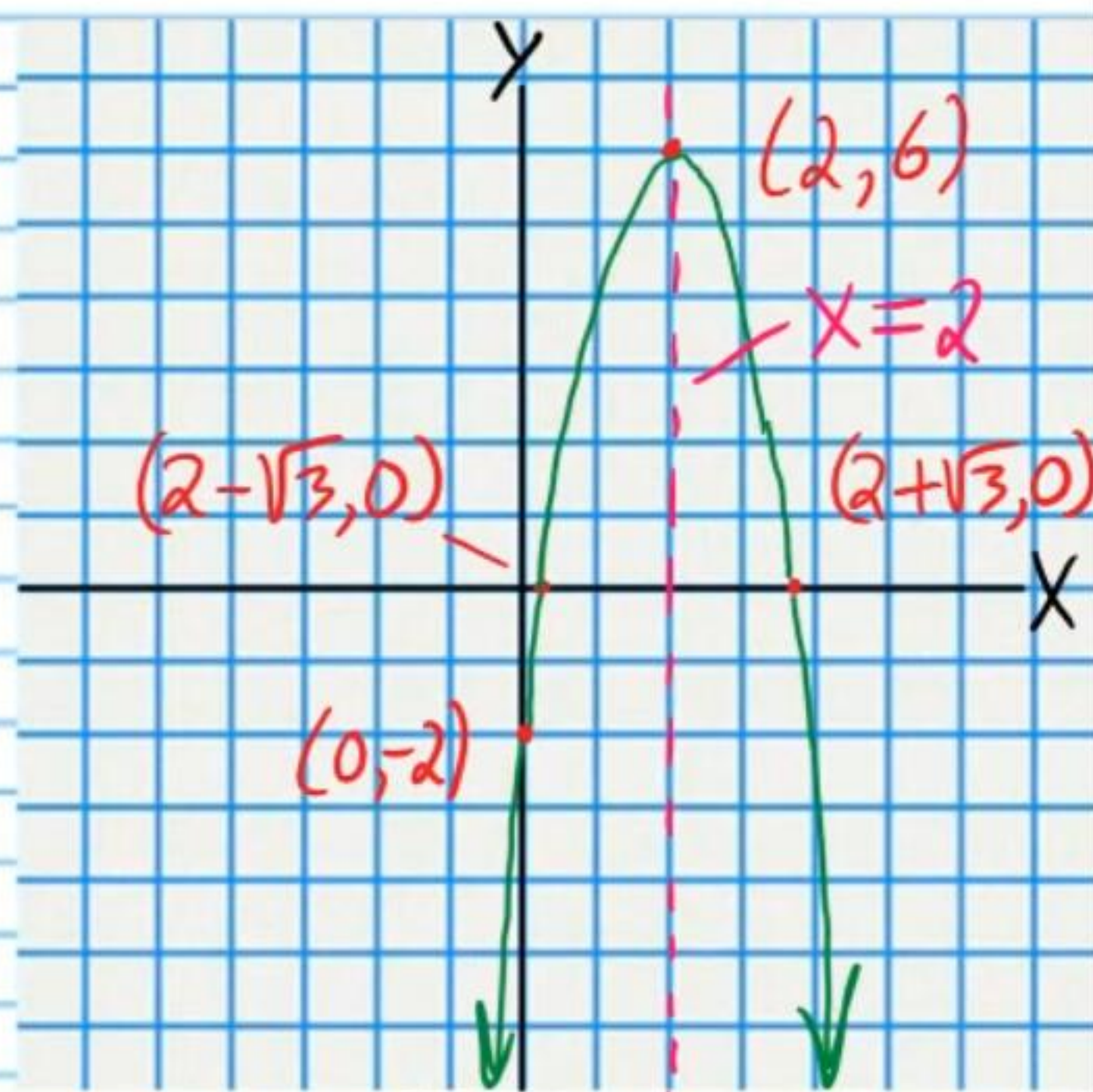
III) X-Intercepts

$$y = 0$$

$$0 = -2(x-2)^2 + 6$$

$$-6 = -2(x-2)^2$$

$$3 = (x-2)^2$$



$$\pm\sqrt{3} = \sqrt{(x-2)^2}$$

$$\pm\sqrt{3} = x-2$$

$$2 \pm \sqrt{3} = x$$

$$x = 2 + \sqrt{3} \text{ or } x = 2 - \sqrt{3}$$

$$(2 + \sqrt{3}, 0) \text{ \& } (2 - \sqrt{3}, 0)$$

$$x \approx 2 + 1.732 \text{ or } x \approx 2 - 1.732$$

$$x \approx 3.732 \text{ or } x \approx 0.268$$

$$\textcircled{3} -4x^2 + 3 = y$$

$$-4(x-0)^2 + 3 = y$$

$$a = -4, h = 0, k = 3$$

I) Vertex

$$V_x = h = 0, V_y = k = 3$$

$$V(0, 3)$$

II) Y-Intercept

$$x = 0$$

$$-4(0)^2 + 3 = y$$

$$-4(0) + 3 = y$$

$$3 = y$$

$$(0, 3)$$

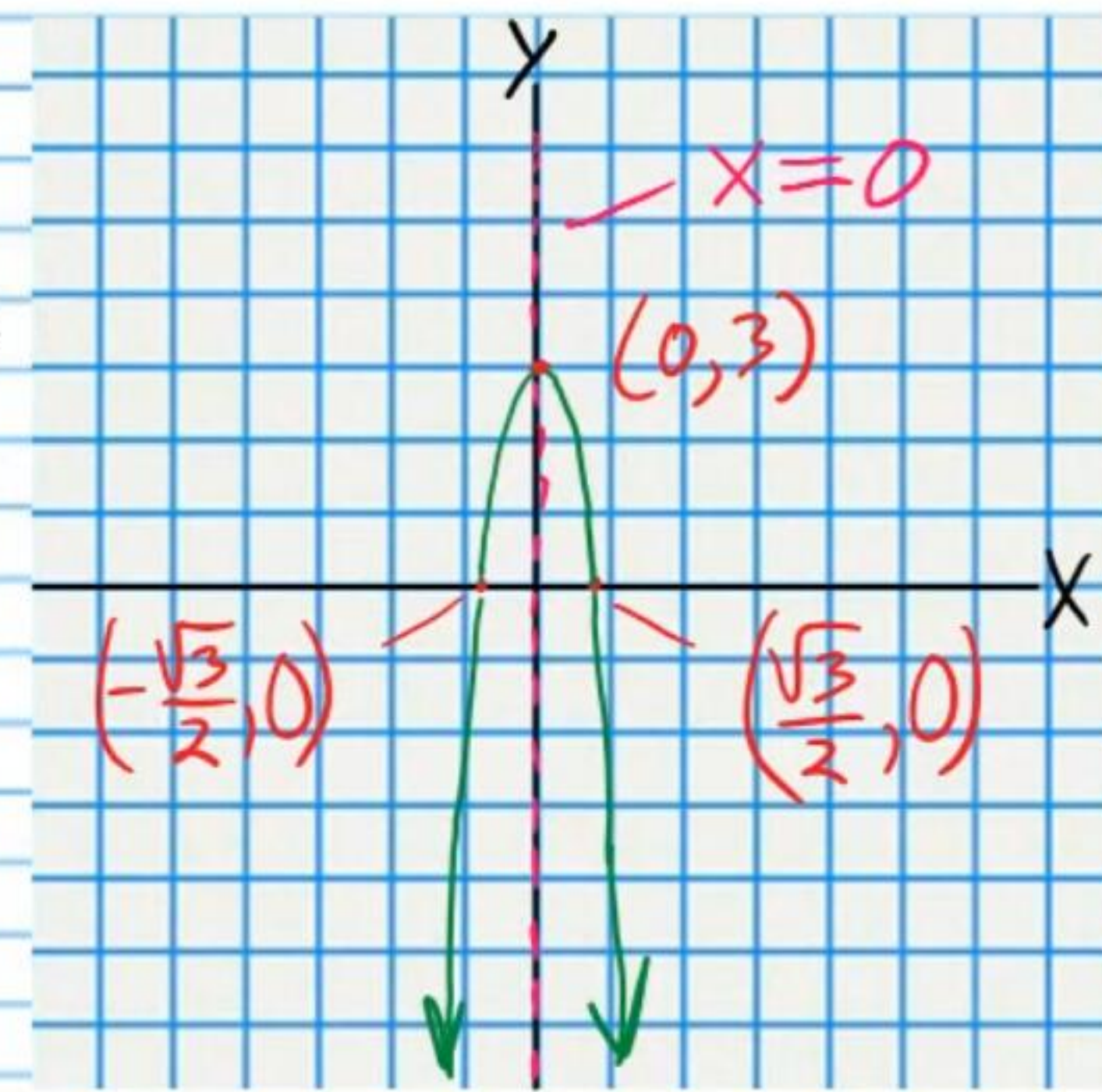
III) X-Intercepts

$$y = 0$$

$$-4x^2 + 3 = 0$$

$$-4x^2 = -3$$

$$x^2 = \frac{3}{4}$$



$$\sqrt{x^2} = \pm\sqrt{\frac{3}{4}}$$

$$x = \pm \frac{\sqrt{3}}{\sqrt{4}}$$

$$x = \pm \frac{\sqrt{3}}{2}$$

$$\left(\frac{\sqrt{3}}{2}, 0\right) \text{ \& } \left(-\frac{\sqrt{3}}{2}, 0\right)$$

$$(0.866, 0) \text{ \& } (-0.866, 0)$$

④ $y = 3(x-1)^2 - 2$

$a=3, h=1, k=-2$

I) Vertex

$V_x = h = 1, V_y = k = -2$

$V(1, -2)$

II) Y-Intercept

$x = 0$

$y = 3(0-1)^2 - 2$

$y = 1$

$(0, 1)$

III) X-Intercepts

$y = 0$

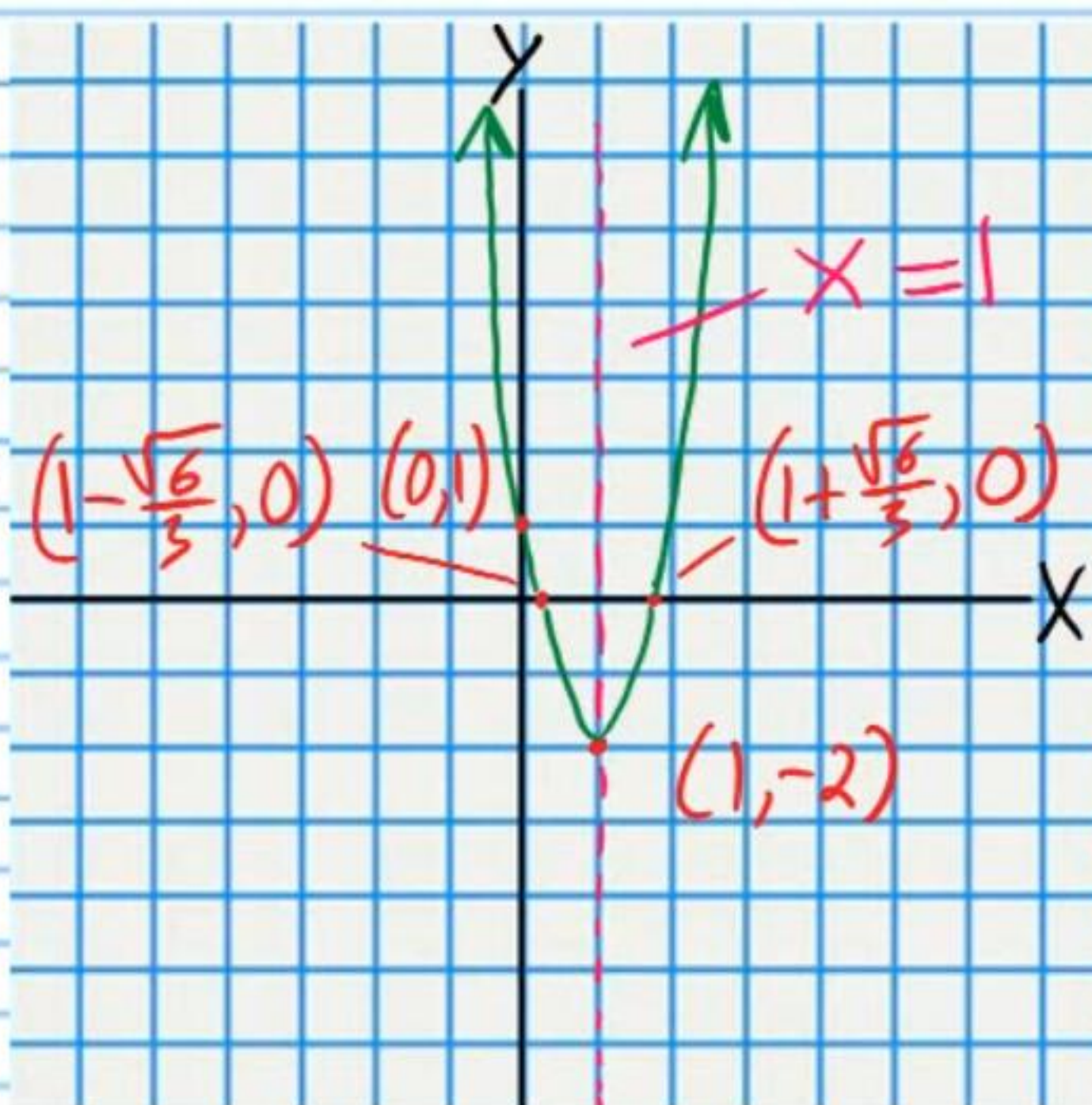
$0 = 3(x-1)^2 - 2$

$2 = 3(x-1)^2$

$\frac{2}{3} = (x-1)^2$

$\pm \sqrt{\frac{2}{3}} = \sqrt{(x-1)^2}$

$\pm \sqrt{\frac{2}{3}} = x-1$



$1 \pm \sqrt{\frac{2}{3}} = x$

$1 \pm \frac{\sqrt{2}}{\sqrt{3}} = x$

$1 \pm \frac{\sqrt{6}}{3} = x$

$x = 1 + \frac{\sqrt{6}}{3}$ or $x = 1 - \frac{\sqrt{6}}{3}$

$(1 + \frac{\sqrt{6}}{3}, 0)$ & $(1 - \frac{\sqrt{6}}{3}, 0)$

$x \approx 1 + 0.816$ or $x \approx 1 - 0.816$

$x \approx 1.816$ or $x \approx 0.183$

Finding the Y-intercept in Standard Form

If $y = ax^2 + bx + c$ is graphed on an xy -plane, then $(0, c)$ is the y -intercept of the graph.

⑤ $y = x^2 - 6x + 5$

$y = (x-1)(x-5)$

$a=1, m=1, n=5$

I) Vertex

$V_x = \frac{m+n}{2} = \frac{1+5}{2}$

$= 3$

$V_y = (3-1)(3-5)$

$V_y = (2)(-2) = -4$

$V(3, -4)$

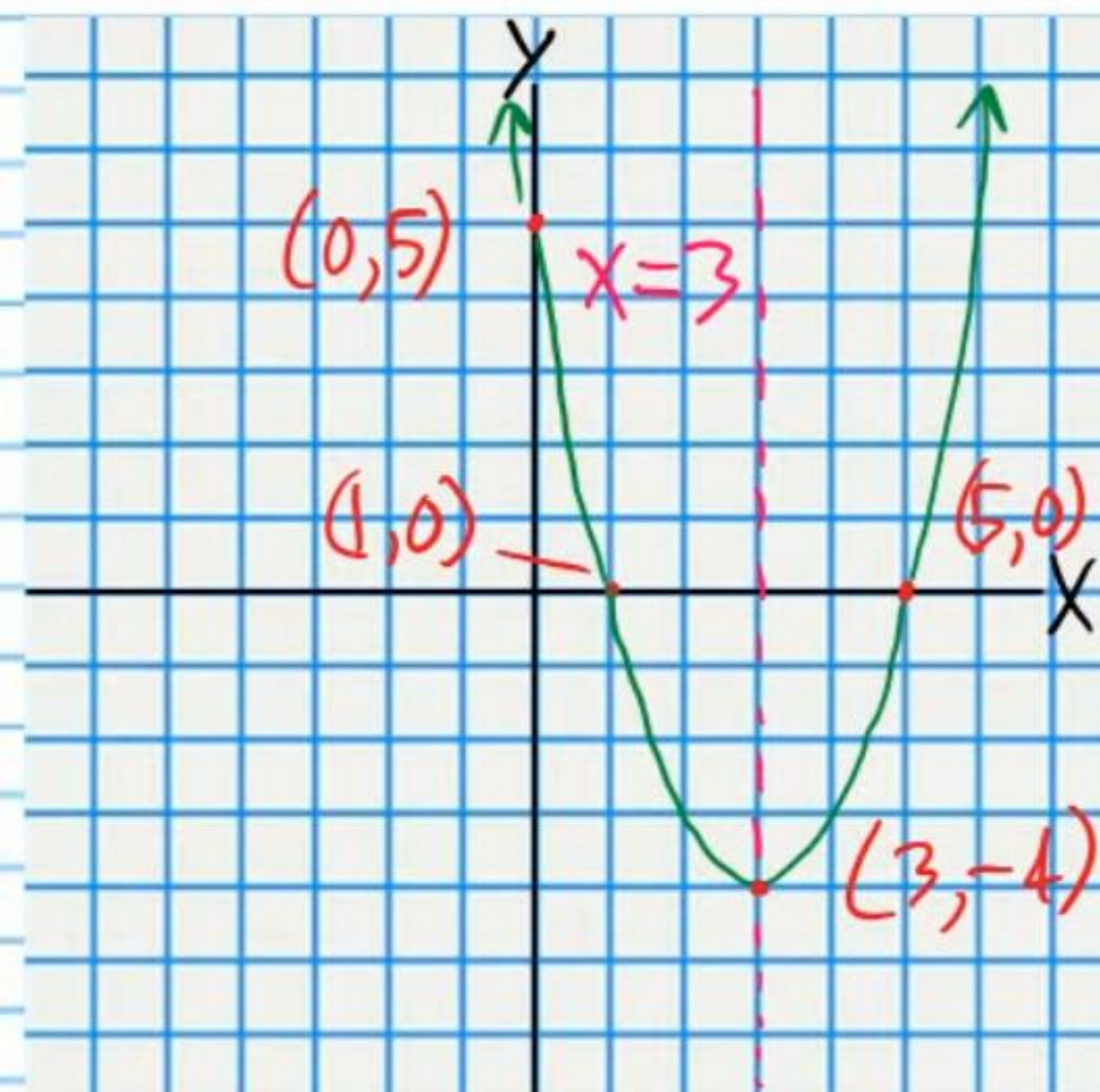
II) Y-Intercept

$x = 0$

$y = 0^2 - 6(0) + 5$

$y = 0 - 0 + 5$

$y = 0 - 0 + 5$



$y = 5$

$(0, 5)$

III) X-Intercepts

$y = 0$

$0 = (x-1)(x-5)$
 $(1, 0)$ & $(5, 0)$

$$\textcircled{6} x^2 + 8x + 15 = y$$

$$(x+5)(x+3) = y$$

$$a=1, m=-5, n=-3$$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{-5+(-3)}{2}$$

$$= -4$$

$$V_y = (-4+5)(-4+3)$$

$$V_y = (1)(-1)$$

$$V_y = -1$$

$$V(-4, -1)$$

II) Y-Intercept

$$x=0$$

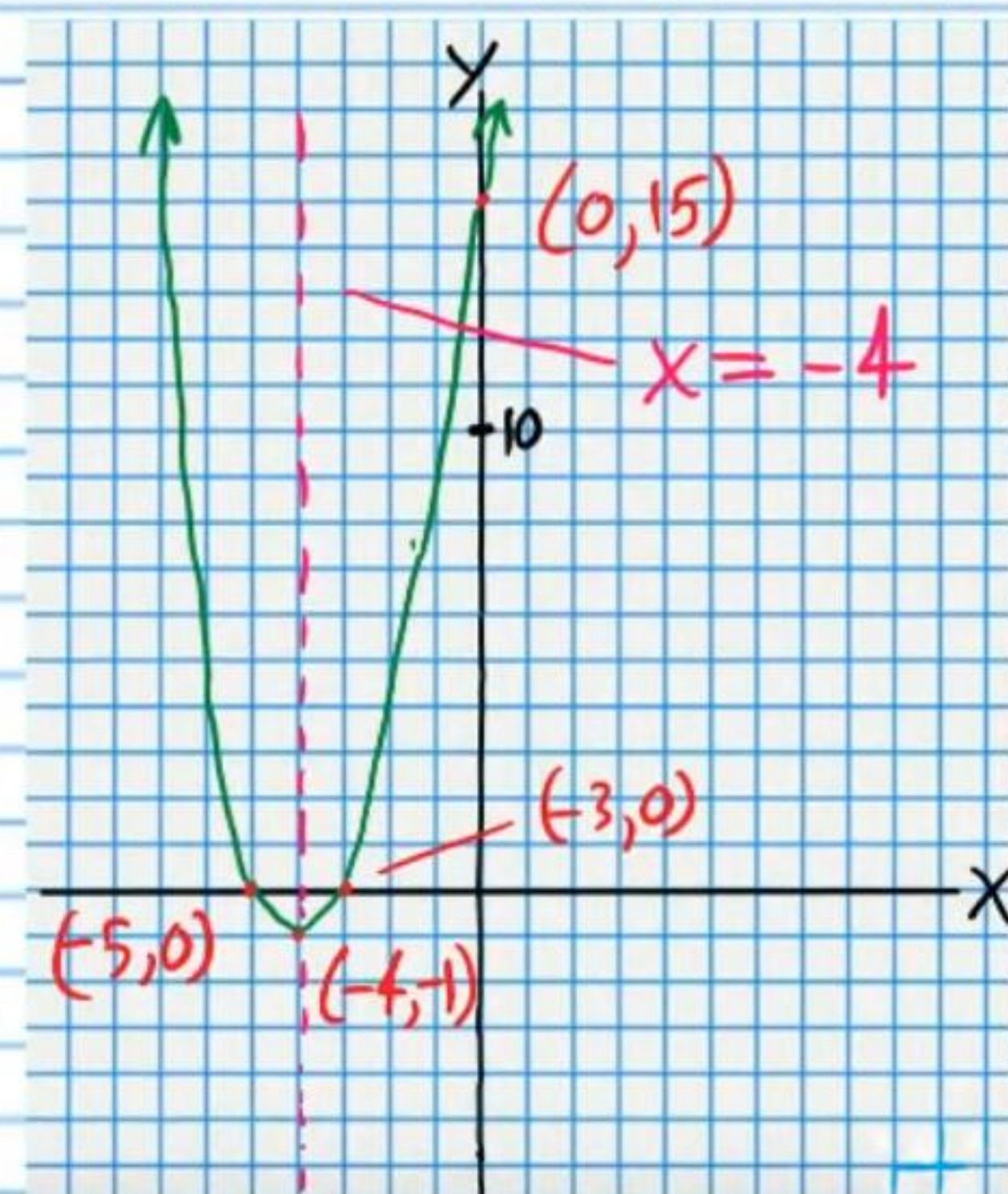
$$y=15$$

$$(0, 15)$$

III) X-Intercepts

$$y=0$$

$$(-5, 0) \& (-3, 0)$$



$$\textcircled{7} x^2 - 2x - 8 = y$$

$$(x-4)(x+2) = y$$

$$a=1, m=4, n=-2$$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{4+(-2)}{2}$$

$$= 1$$

$$V_y = (1-4)(1+2)$$

$$V_y = (-3)(3) = -9$$

$$V(1, -9)$$

II) Y-Intercept

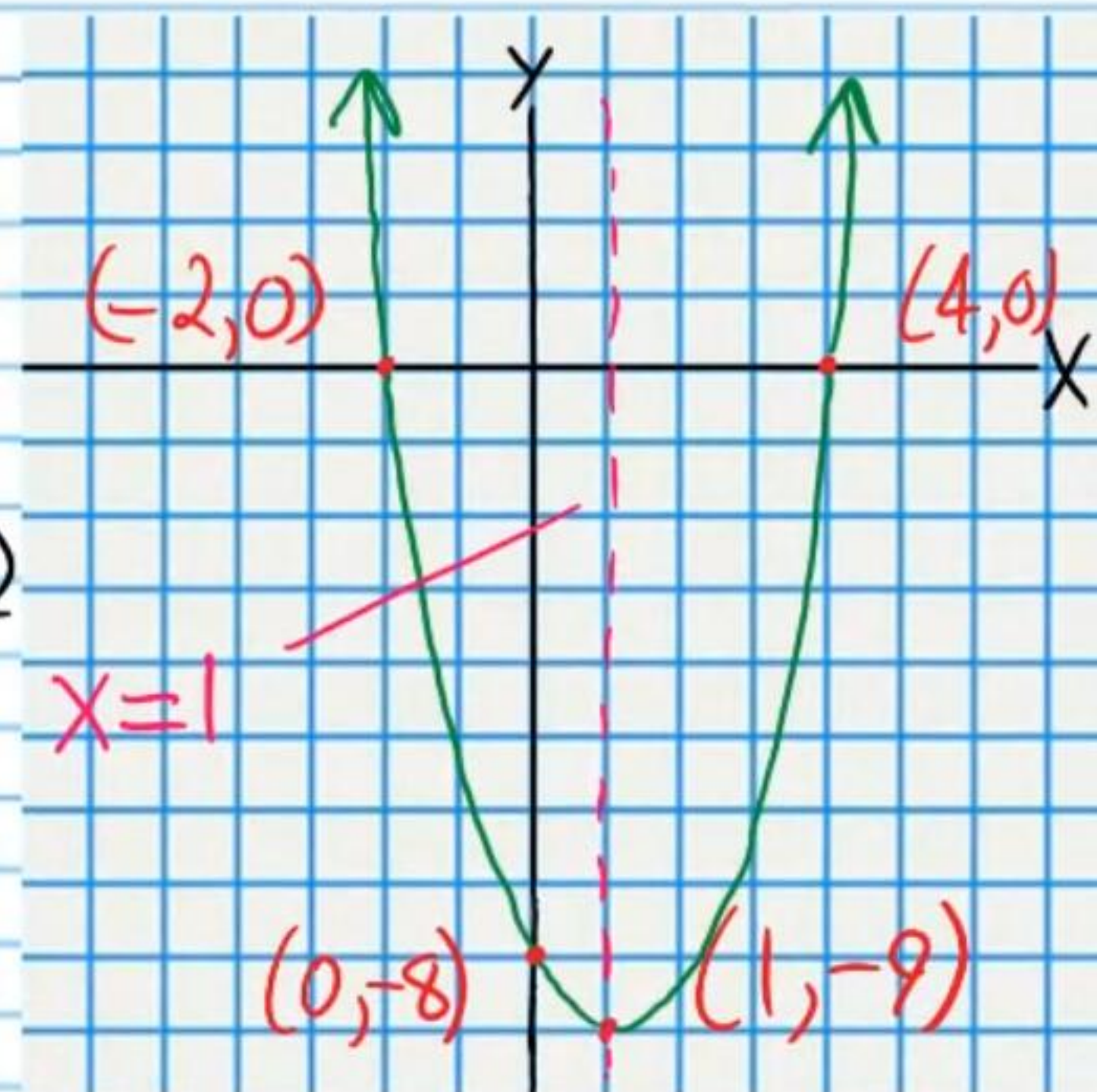
$$x=0$$

$$(0, -8)$$

III) X-Intercepts

$$y=0$$

$$(4, 0) \& (-2, 0)$$



⑧ $y = x^2 + 6x - 7$

$y = (x+7)(x-1)$

$a=1, m=-7, n=1$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{-7+1}{2}$$

$$= -3$$

$V_y = (-3+7)(-3-1)$

$V_y = (4)(-4) = -16$

$V(-3, -16)$

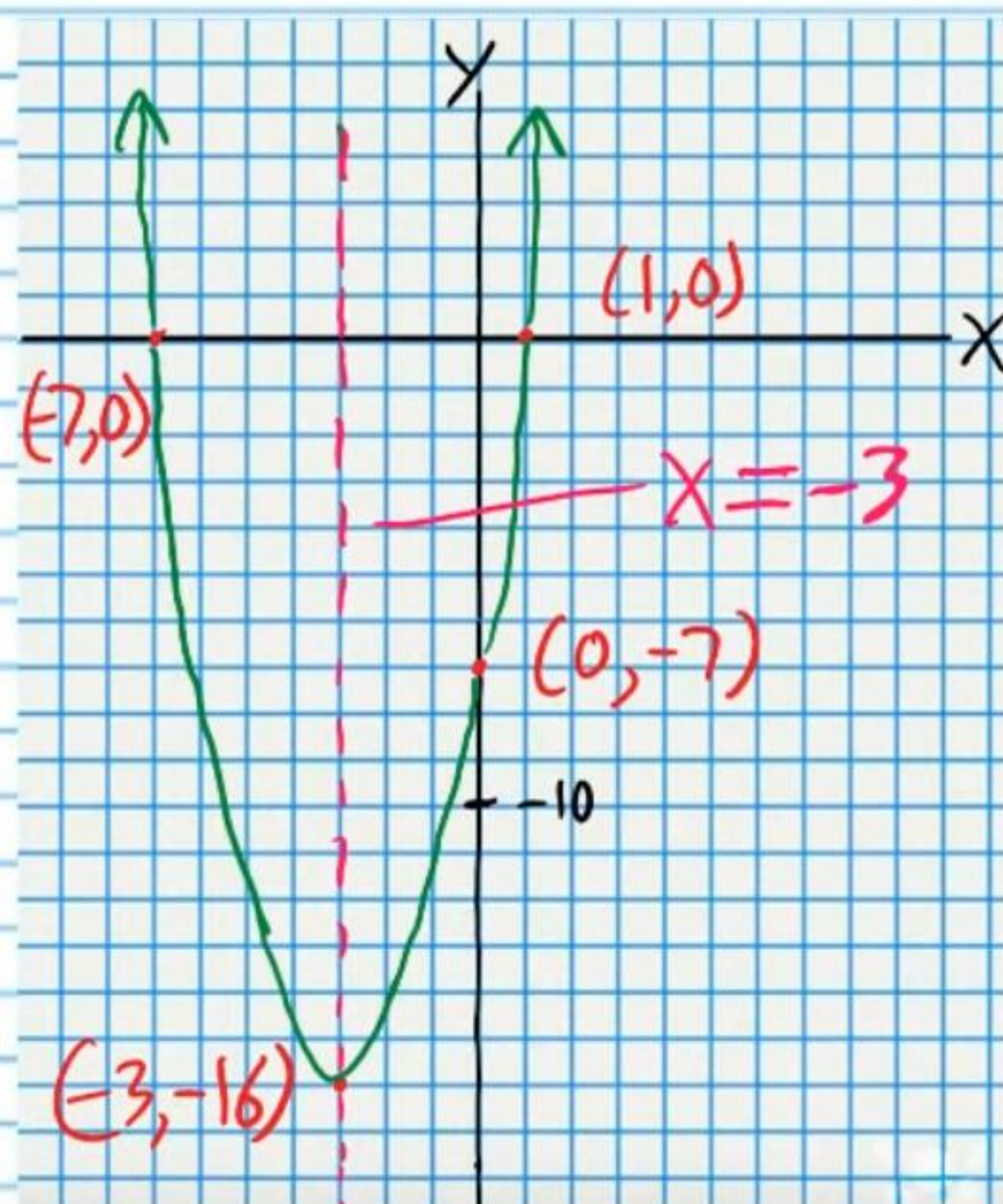
II) Y-Intercept

$(0, -7)$

III) X-Intercepts

$y=0$

$(-7, 0) \& (1, 0)$



⑨ $y = -x^2 - 4x + 12$

$y = -1(x^2 + 4x - 12)$

$y = -1(x+6)(x-2)$

$y = -(x+6)(x-2)$

$a=-1, m=-6, n=2$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{-6+2}{2}$$

$$= -2$$

$V_y = -(-2+6)(-2-2)$

$V_y = -(4)(-4)$

$V_y = 16$

$V(-2, 16)$

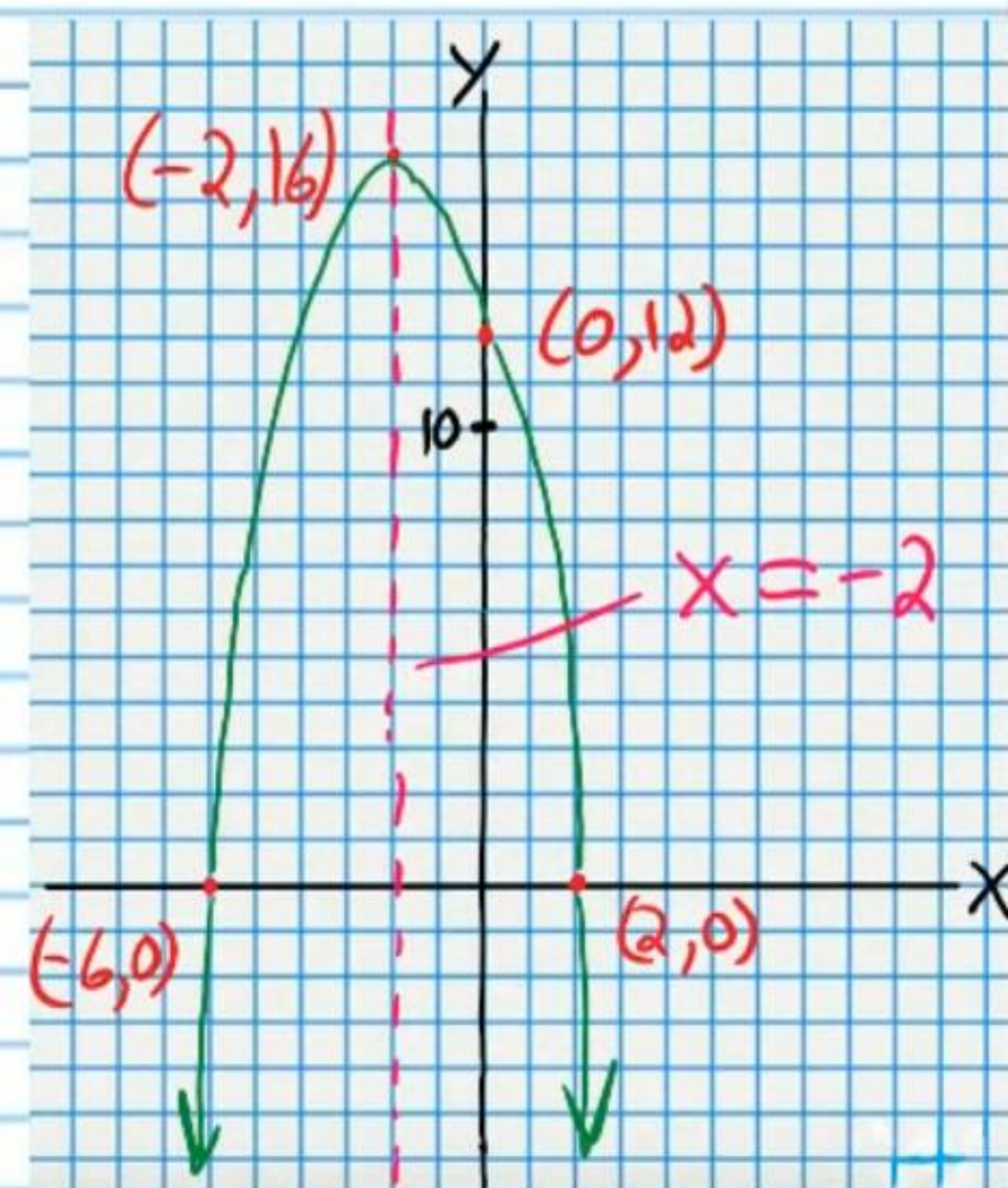
II) Y-Intercept

$x=0$

$(0, 12)$

III) X-Intercepts

$(-6, 0) \& (2, 0)$



$$\textcircled{10} y = -x^2 - 8x - 15$$

$$y = -1(x^2 + 8x + 15)$$

$$y = -1(x+5)(x+3)$$

$$y = -(x+5)(x+3)$$

$$a = -1, m = -5, n = -3$$

I) Vertex

$$V_x = \frac{m+n}{2} = \frac{-5+(-3)}{2}$$

$$= -4$$

$$V_y = -(-4+5)(-4+3)$$

$$V_y = -(1)(-1)$$

$$V_y = 1$$

$$V(-4, 1)$$

II) Y-Intercept

$$x = 0$$

$$(0, -15)$$

III) X-Intercepts

$$(-5, 0) \text{ \& } (-3, 0)$$

